AI / ML / Data Salary Prediction

A comprehensive data product to predict salary metrics using Machine Learning models.

Diana Yule · Bibian

Ironhack Data Analytics Bootcamp — Week 7

Project Overview

Goal

As a students of DA bootcamp we are interested in estimating the money opportunities that the job market could offer after this bootcamp.
Thus, this project aims to construct a model that could **predict the salary_in_usd for global AI / ML / Data roles** using supervised regression models.

Data

aijobs.net · ~34k rows · CC0 licence
Direct CSV download.

Team Split

Model 1 ⟶ company-side features (job title, company size, location, year)

Model 2 ⟶ experience-side features (experience level, employment type…)

🤝 Team Split

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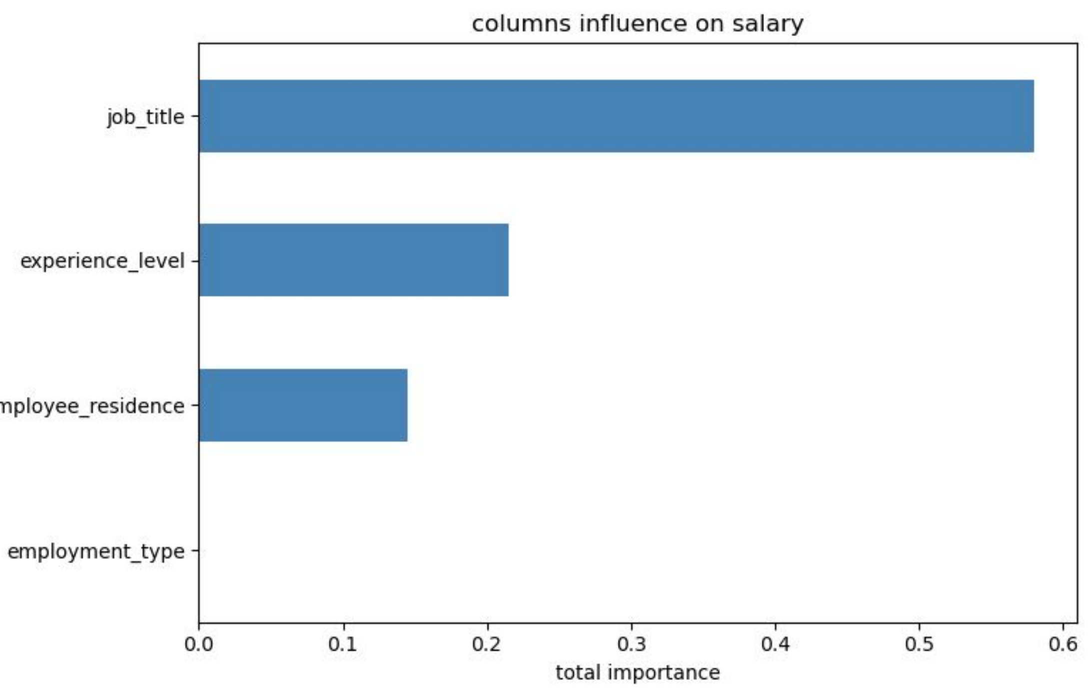


Data Selection & Preparation

~34k

rows in raw dataset

29,562

rows after deduplication

80 / 20

train / test split

1. Source — aijobs.net — direct CSV download (CC0 licence).

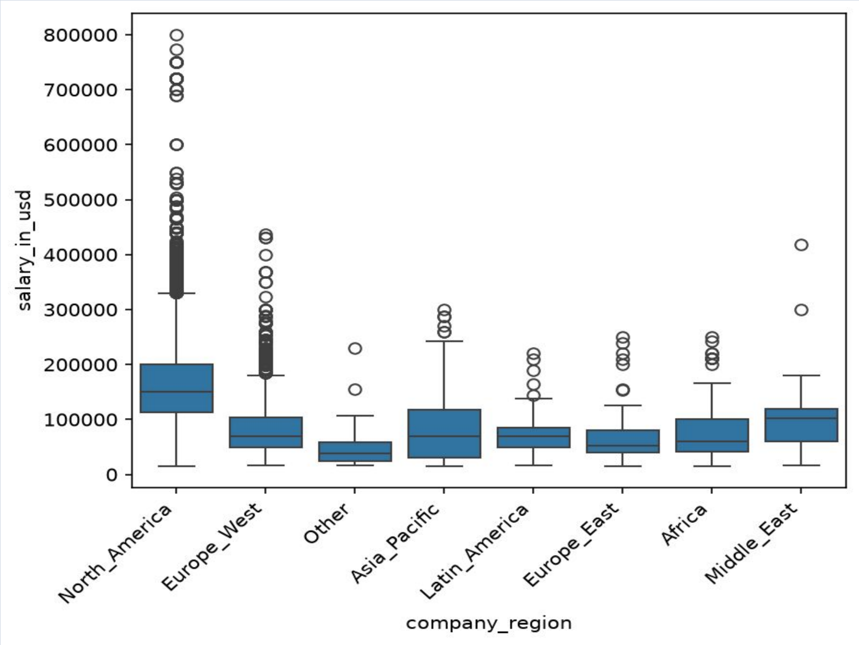
2. Drop leakage — Removed raw local-currency salary column — would trivially leak the target.

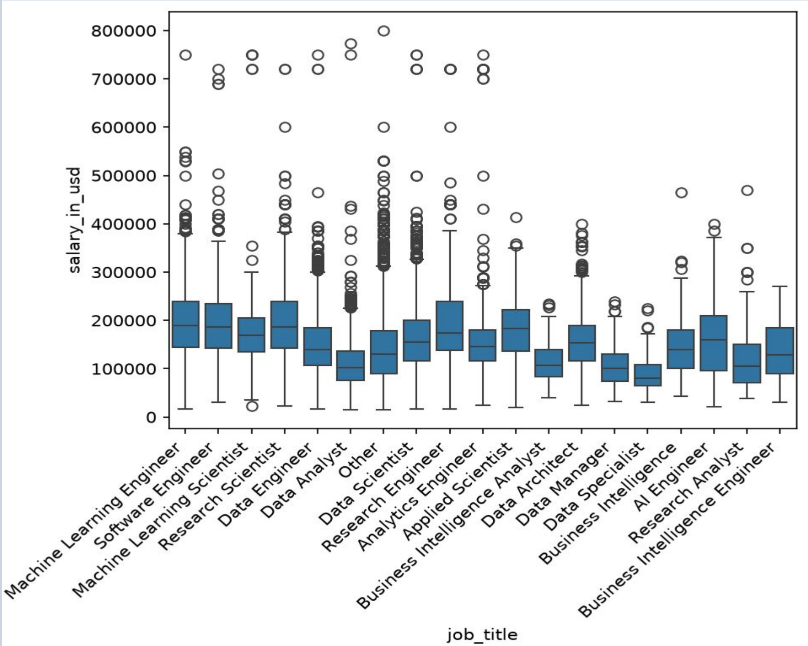
3. Nulls / types — No nulls; all columns in correct dtype after load.

4. Outlier check — Target (salary_in_usd) heavily right-skewed; kept all rows — no hard cap applied.

5. Split — Stratified 80/20 split on work_year to preserve year distribution.

Categorical data





North America dominates salary ranges. The **regional grouping reduces the high-cardinality country** column from 100+ categories to 7.
There are around **196 job titles**. As many are uncommon, we **filter** by titles with at least **150 occurrences.**

Model 1 - Feature Engineering & Selection

Diana's company-side feature set

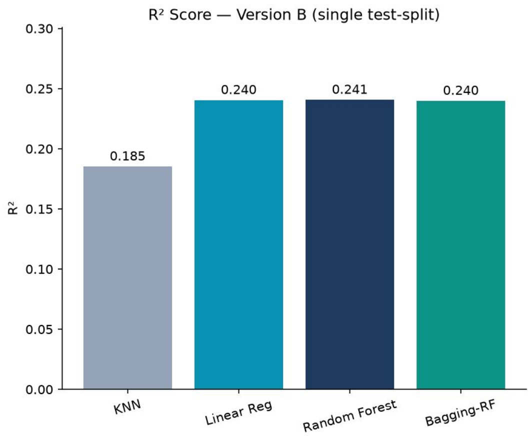
📊 Feature Selection — ANOVA (p)

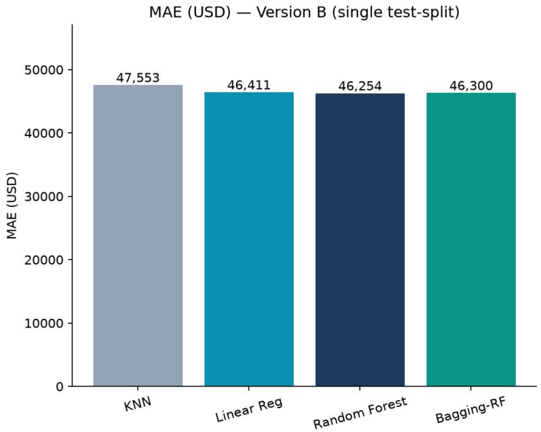
Feature	p	Kept?
job_title (grouped)	0.0	✅
company_region	0.0	✅
company_size	5.7e-78	✅

Does the mean salary_in_usd is equal across all categories of the column → the column carries no salary signal → candidate to drop.

We **reject H0 f**or all three columns (p < 0.05), so each explains some salary variance — we keep all categorical columns.

🔧 Choosing a model Baseline





Different models baselines - single spit

Model 2 - Feature Engineering & Selection

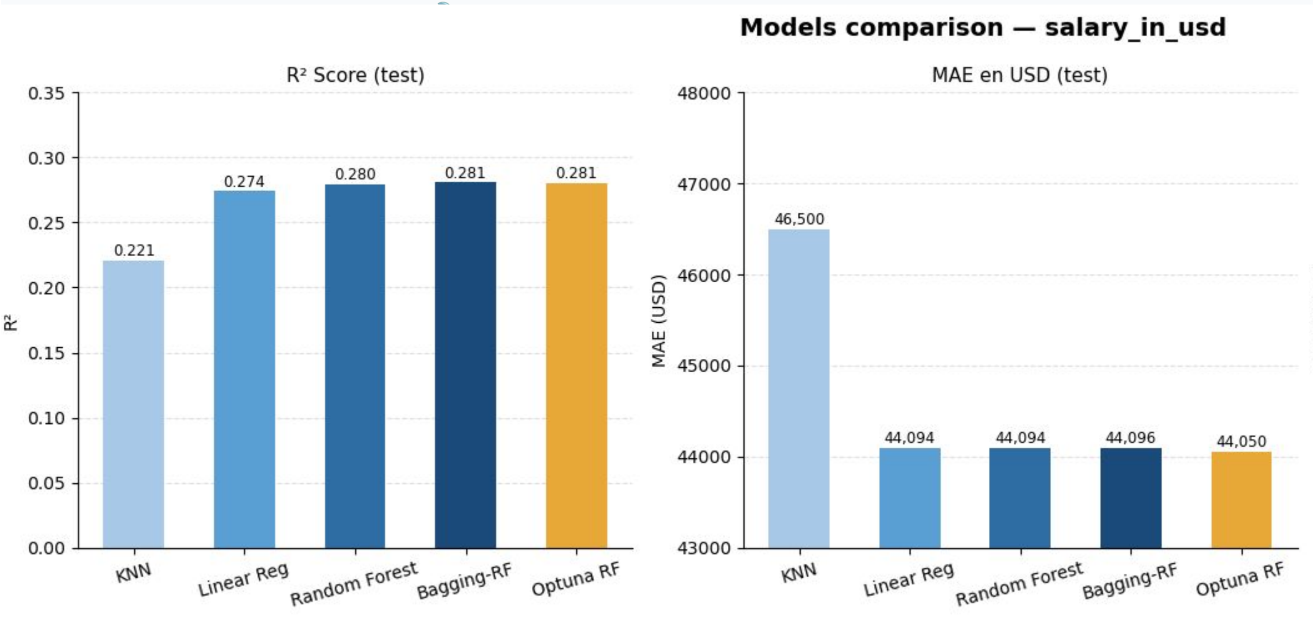
Bibian's company-side feature set

Feature Selection — ANOVA (p)

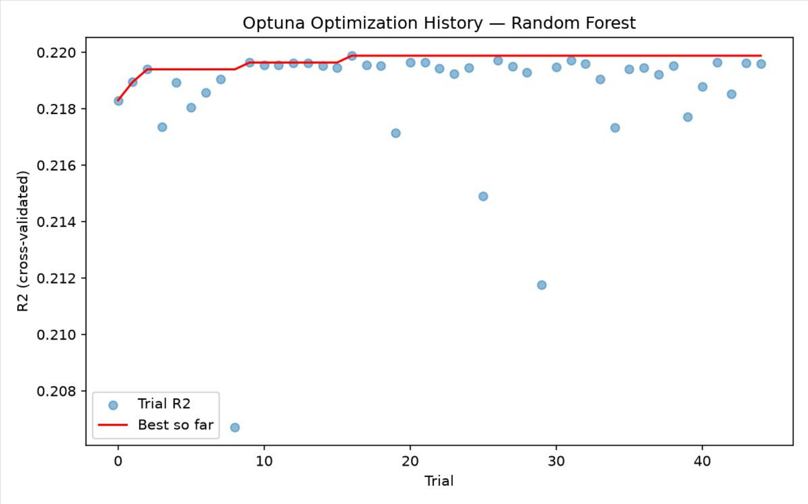
Feature	p	Kept?
experience_lev	0.0	✓
mployment_ty	0.0	✓
job_title	0.0	✓
nployee_reside	0.0	✓

Does the mean salary_in_usd is equal across all categories of the column → the column carries no salary signal → candidate to drop.

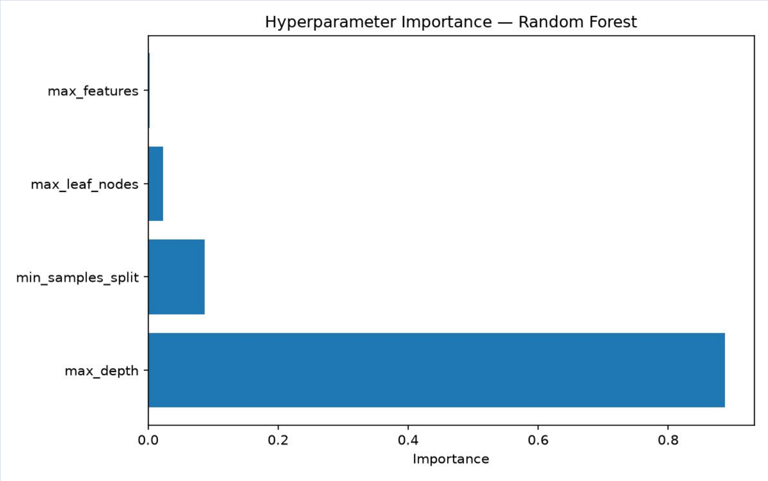
We reject H0 for all columns ($p < 0.05$), so each explains some salary variance — we keep all categorical columns.



Model 1 - Hyperparameter Tuning — Bayesian Optimisation (Optuna)



Optimisation History (45 trials)

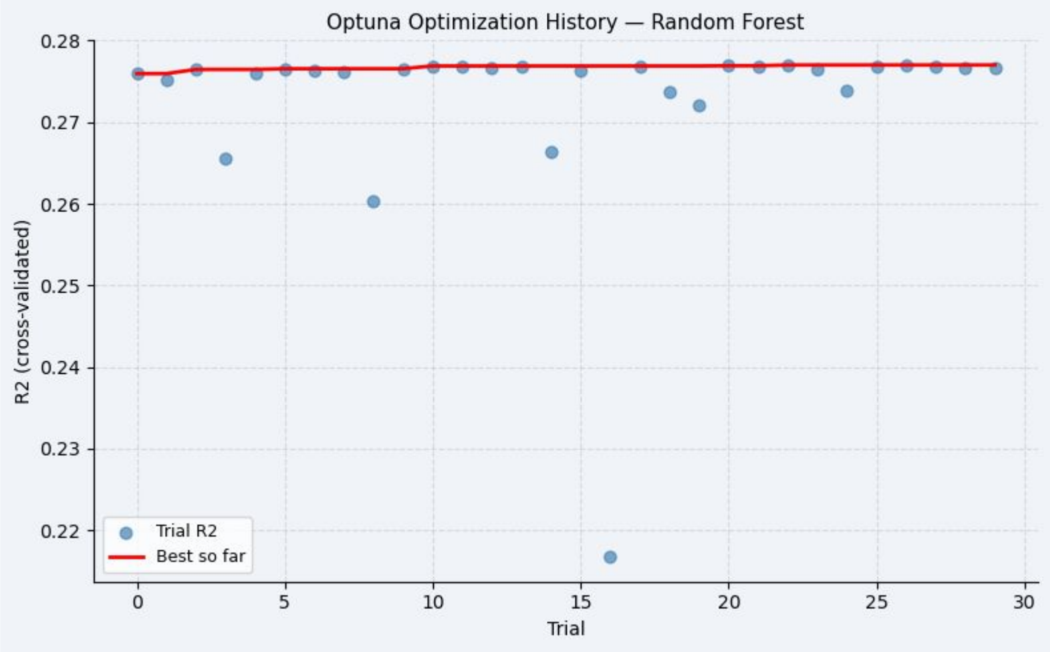


Hyperparameter Importance

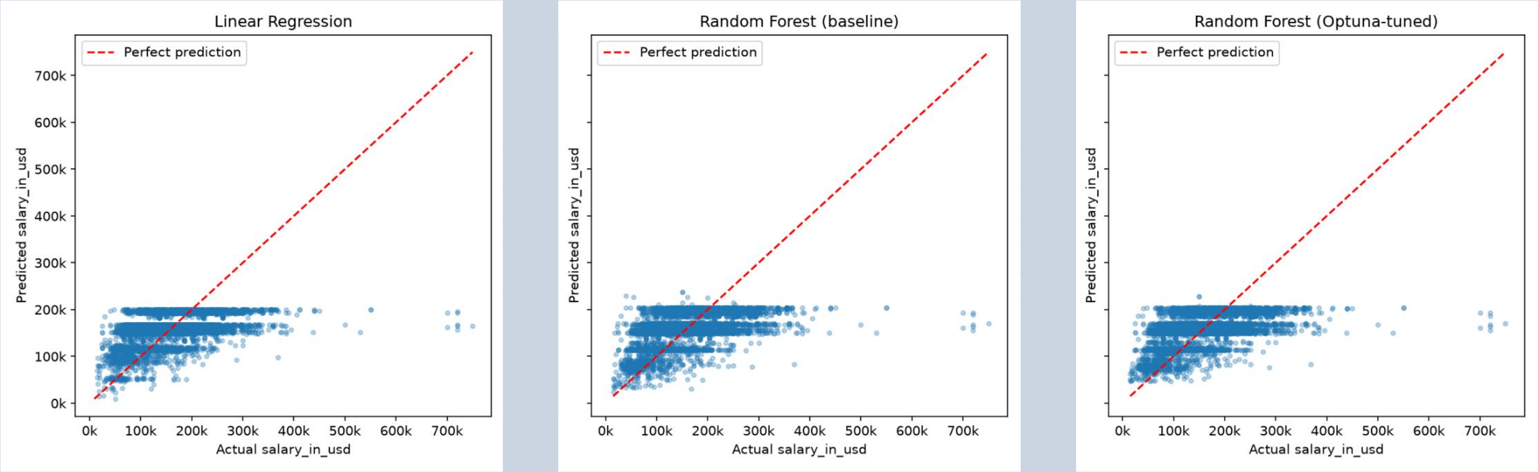
Sampler: 10-fold CV objective · Tuned: max_depth, min_samples_split, max_leaf_nodes, max_features.
Mean R2: 0.2171

The R2 with 95% confidence interval for the best combination of hyperparameters:: (0.2068, 0.2199, 0.2329)

Model 2 - Bayesian Optimisation (Optuna)



Model 1 - Evaluation

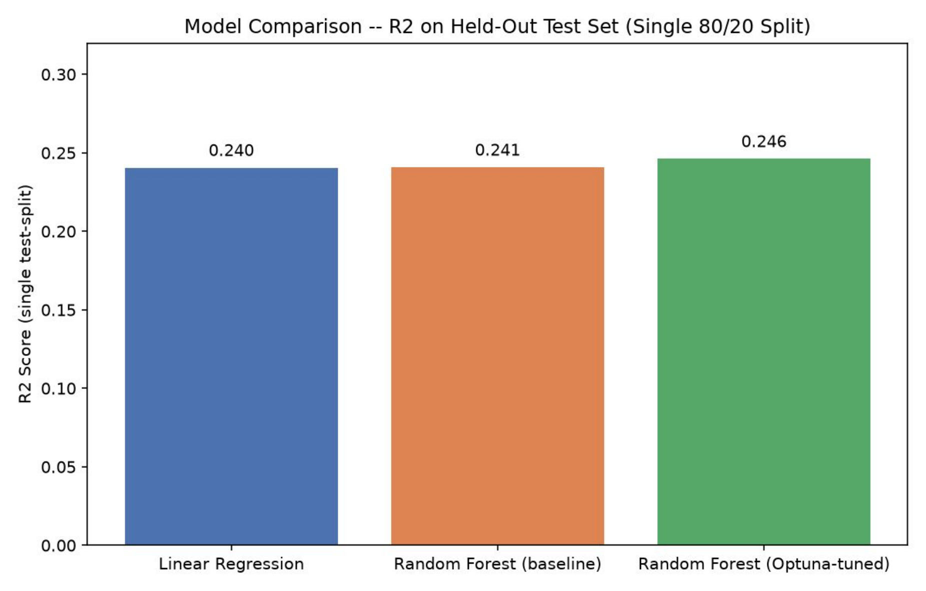


Key Findings

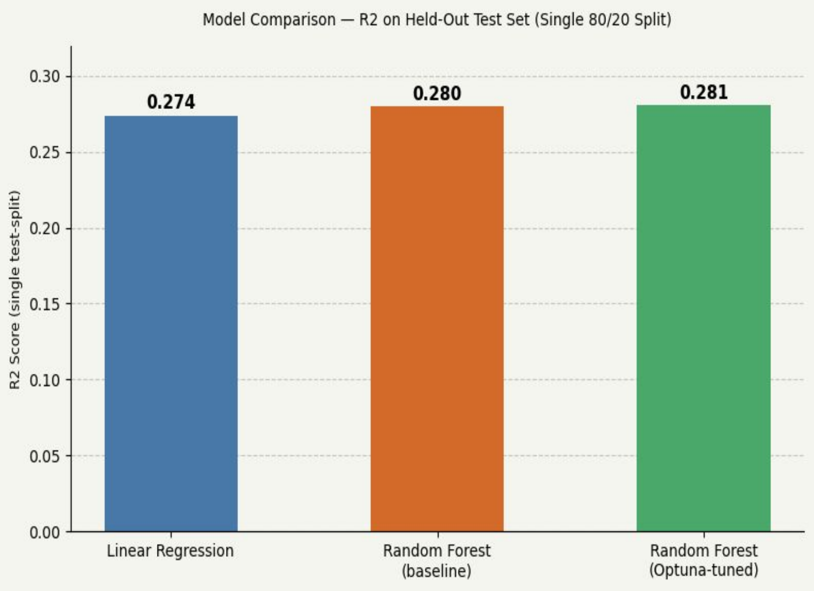
- All 3 model families converge: $R^2 \approx 0.24–0.25$
- LinReg, RandomForest, RF-optimised, all reach same ceiling
- Optuna gain is within noise (~0.003 R^2). Confirms the problem is feature-ceiling, not tuning.
- Structural ceiling — not a tuning problem

Models Evaluation

Model 1: company-side features



Model 2: experience-side features



Key Findings

- Model 1 families convergence ($R^2 \approx 0.24$–0.25) is lower than Model 2 families convergence ($R^2 \approx 0.27$–0.28)
- RF-optimised improvement on model 1 (0.005) was better than model 2 by 4 points.
- For both models, Optuna gain is within noise indicating that the problem is feature-ceiling, not tuning.

Key Findings & Insights

🌍 **Geography matters** By grouping countries in regions we gain speed on the model to the cost of losing countries differences.

📉 **Feature ceiling is real** For all model families tried with Model 1, and Model 2, there is convergence ceiling to $R^2 \approx 0.24, 0.28$ across structurally different algorithms is diagnostic — not a model choice issue. A better feature set matters more than better tuning.

🔧 **Tuning ≠ cure** Despite using Bayesian optimisation for both models the gains are minimal up to ~0.005 R^2.
When you're at a feature ceiling, tuning delivers diminishing returns.

🧪 **R^2 reporting discipline** Single test-split and CV results reported separately. Mixing them inflates apparent performance by 2–3 pp.

Challenges & Learnings

⚠️ KNN Classifier → Regressor

The dummification of many categories and mixing scales among the categories results in more complexity not worth as turned out into less performance.

🔄 Feature engineering before split

Due to the time pressure, a proper dataset split for each model building based on the categorical columns was done intuitively (company fact vs. experience level). For a better approach more time should be allocated to feature selection per model.

🔬 Convergence = diagnosis

Four structurally different algorithms producing the same R^2 is informative — it's not a failure, it's a signal about the feature space.

Real-World Application & Impact

Salary Benchmarking

HR teams can use a tuned model to flag underpaid roles or benchmark new hires against market ranges — especially valuable in remote-first AI orgs where geography previously anchored pay.

Career Navigation

Job-seekers can estimate the USD salary premium of a title, region, or company-size choice before accepting an offer — actionable beyond simply 'get more experience'.

Current & future salary expectancy

One would like to see what is the expected salary in the coming years, however since our dataset predicts only with the years in the train set (2020-2024) current and future salary predictions are not possible, for now with the ML techniques seen.

Thank You

Questions?

Dataset: aijobs.net/salaries/download/ (CC0) | **Diana Yule · Bibian**

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